

INNOVATIVE PRACTICES – ACTIVE LEARNING TECHNIQUES

Degree, Semester & Branch: V Semester B.E. ECE A & B

Course Code & Title: EC3551 Transmission Lines and RF Systems

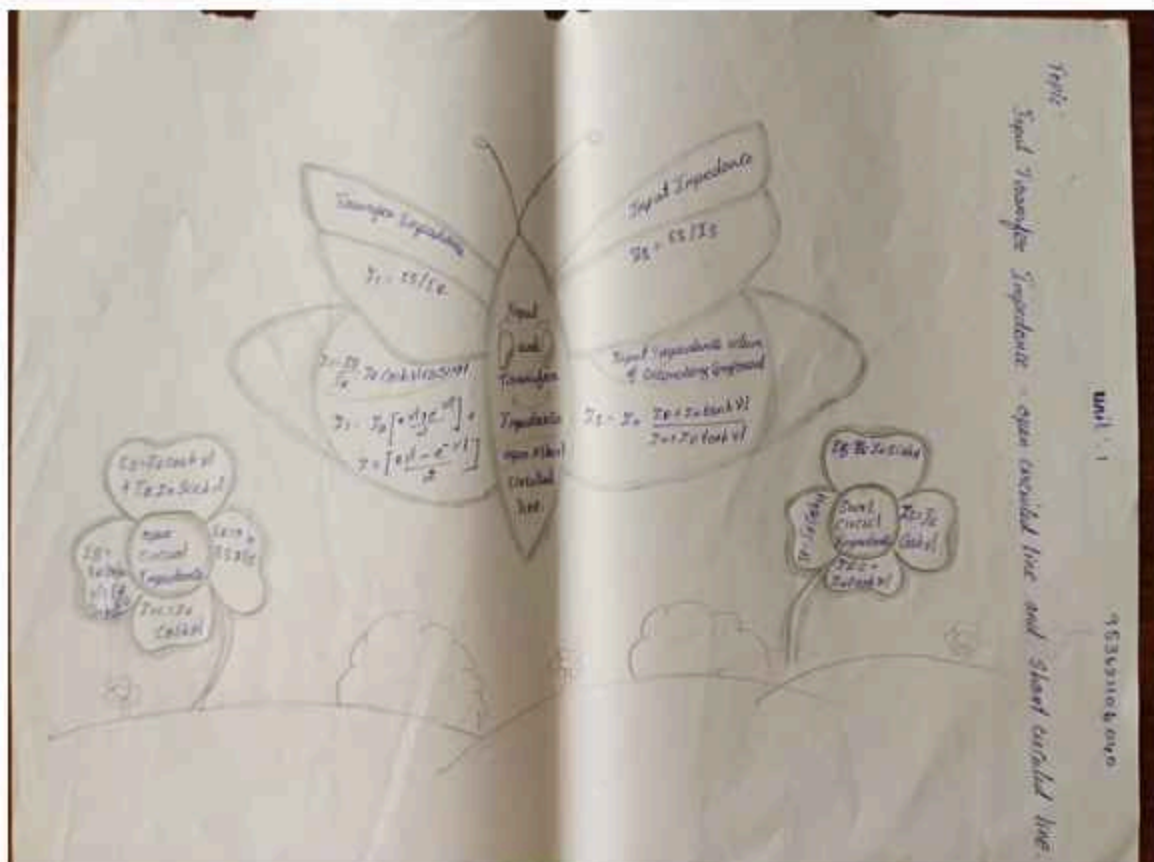
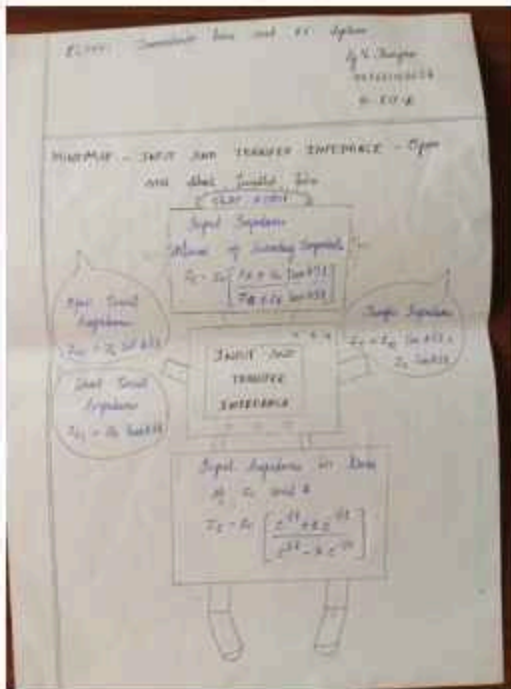
Name of the Faculty member: Mrs.S.Jeeva

UNIT I TRANSMISSION LINE THEORY

Activity: Mind Map

Topic: Input and Transfer Impedance

Date: 05.08.2023



RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

INNOVATIVE TEACHING METHOD-ACTIVE LEARNING TECHNIQUES

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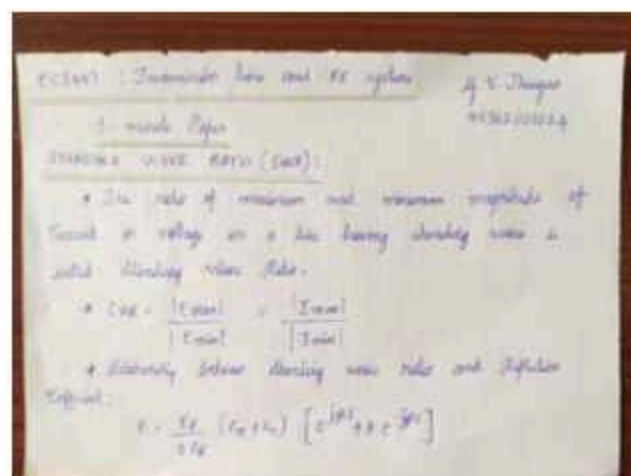
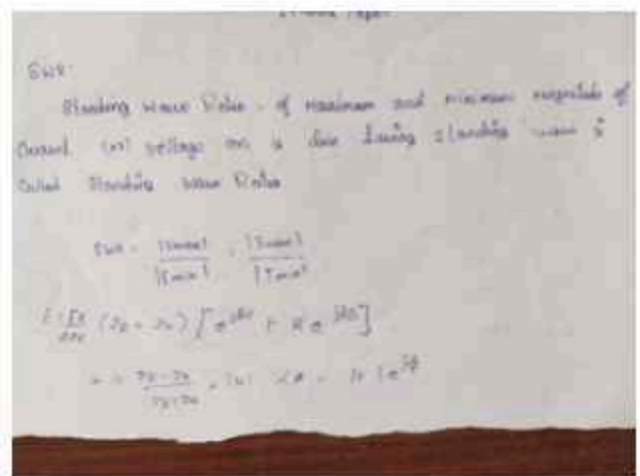
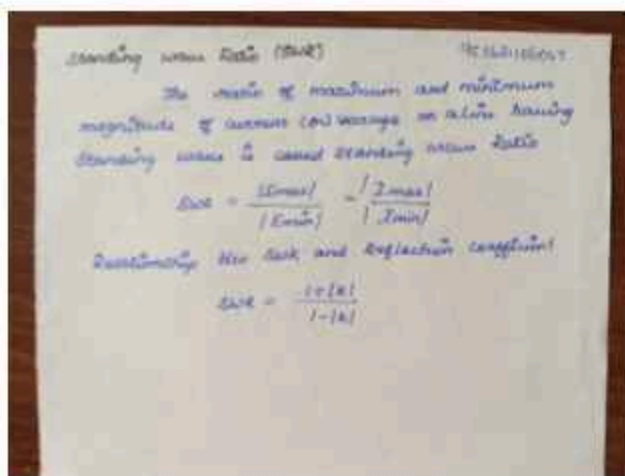
Name of the Faculty member: Mrs.S.Jeeva

UNIT II HIGH FREQUENCY TRANSMISSION LINES

Activity: One Minute Paper

Topic: Standing Wave Ratio

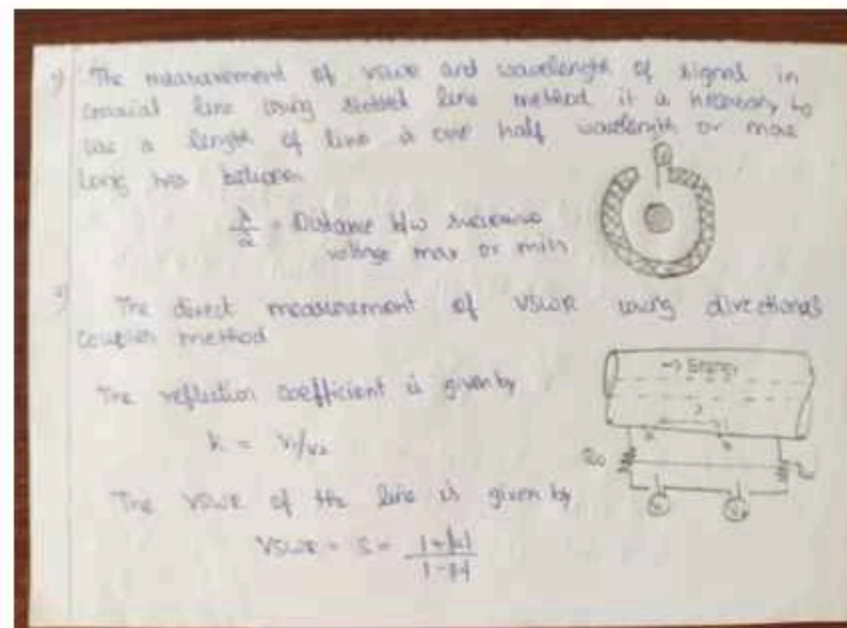
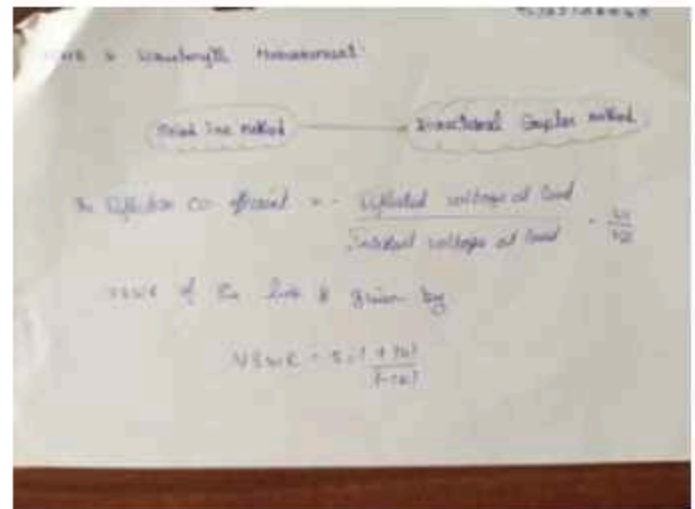
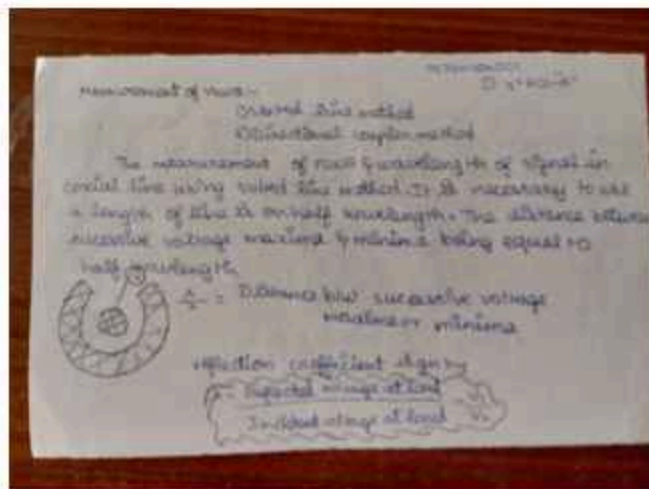
Date: 17.08.2023 & 19.08.2023



Activity: Exit Slip

Topic: Measurement of VSWR and Wavelength

Date: 24.08.2023 & 24.08.2023



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UNIT III IMPEDANCE MATCHING IN HIGH FREQUENCY LINES

Activity: Brain Storming

Topic: Impedance Matching

Date: 26.08.2023 & 31.08.2023



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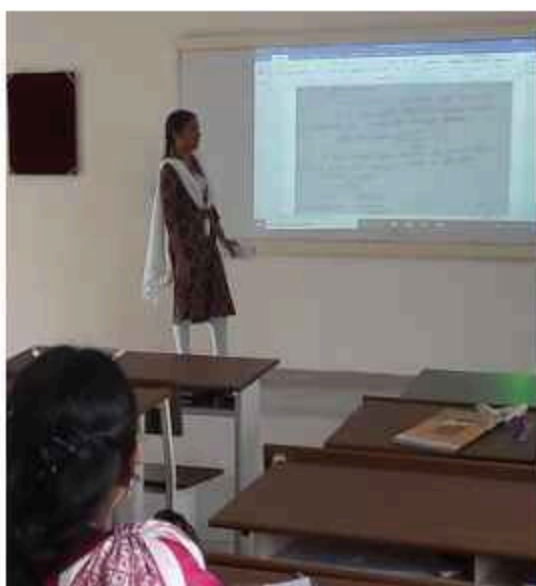
Name of the Faculty member: Mrs.S.Jeeva

UNIT IV WAVEGUIDES

Activity: Zero Minute Speech

Topic: TM Waves in Circular Waveguide

Date: 19.10.2023





Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A & B

Course Code & Title: EC3551 Transmission Lines and RF Systems

Name of the Faculty member: Mrs.S.Jeeva

Innovative Practice Description

- **Unit / Topic:** Unit III – Single Stub Matching
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Collaborative Learning- STAD
- **Justification:**

Student Teams-Achievement Divisions (STAD) is a cooperative learning strategy used in education. It is designed to promote positive interdependence among students within a group, encouraging them to work together to achieve common learning goals..

Time Allotted for the Activity: 45 minutes

• **Details of the Implementation:**

- **Formation of Heterogeneous Groups:** Students are divided into small groups, usually consisting of four to six members. These groups are intentionally heterogeneous, meaning they include students with a range of abilities and backgrounds.
- **Introduction of Learning Material:** The teacher introduces a new concept (Procedure for solving Single stub Matching using Smith Chart) or learning material to the whole class.
- **Individual Study and Preparation:** Each student in the group is responsible for understanding and solving the problem individually. They may work on Problem on their own.

- **Assessments:** After individual preparation, students come together within their groups to solve the another problem or discuss the correct answer for previous problem. This step encourages communication and helps students clarify their problem solving capability.
- **Group Recognition and Rewards:** Group members are collectively responsible for ensuring that each member understands the Problem.

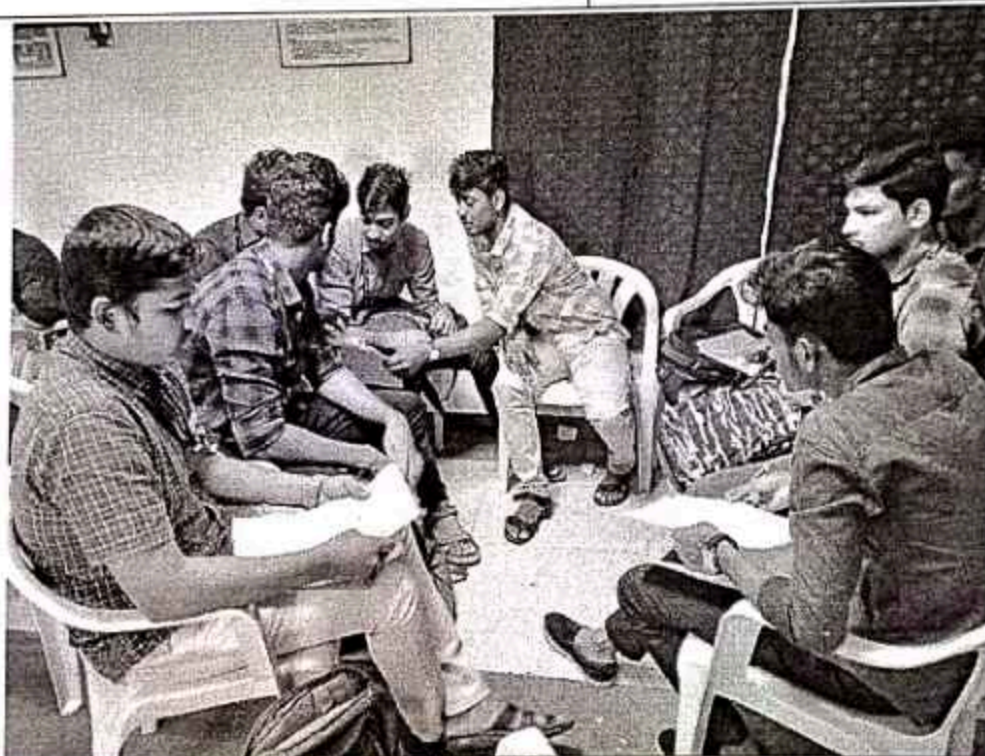
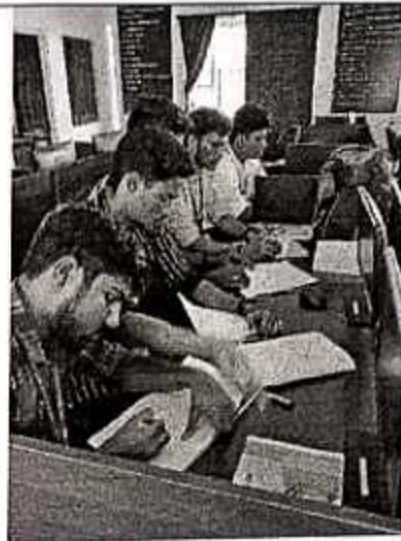
CO – PO / PSO mapping:

CO	PO1	PO8	PO9	PO10	PSO3
CO3	2	2	2	3	3

• PO / PSO mapped:

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	2	2	3	3
Justification for correlation	Smith chart problem requires a basic engineering fundamentals such as rectangular to polar and polar to rectangular conversion, wavelength & frequency relationship. So this course outcome is highly correlated	STAD activity involves group discussion which apply ethical principles to solve the problem. The course outcome is moderately correlated	As the activity involves team work, the leadership qualities such as implementing norms of practice as a team to achieve a goal, listening to team members and treating them equally the Course outcome is mapped with moderate correlation	Through these activity students can communicate the technical and non-technical information with their team members effectively. Hence the course outcome is mapped with high correlation	It is mapped with correlation level 3. Because to develop o analyze transmission line required fo a communication system, knowledge about the parameters, Position & length of the stub is required Hence thi activity i mapped with high correlator

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

Every student is actively involved in the learning process. Since each member of the group plays a crucial role in contributing to the overall understanding of the topic, there is a higher level of engagement. Students enjoy the challenge of working together to share the knowledge among them. Students appreciate the opportunity to learn from their peers.

- ❖ *Benefit of the practice:*

Every students got equal opportunity to come forward to take part in this activity. Students easily understand the single stub matching concepts and solve the problem by using smith chart. This practice helped the students recall the formulae and solve the problem correctly. Students who formed groups were better able to understand concepts by exchanging ideas with one another.

- ❖ *Challenges faced in implementation:*

- The main challenge faced is group conflict. Few Students are not ready to work together. Initially, the students not came up to involve in this activity voluntarily.
- Faced issues while make the students to understand about STAD activity rules.
- This task has been scheduled for 45 minutes. Forming a group takes some time for the students.

References:

TEXTBOOKS

1. John D Ryder, "Networks lines and fields", Prentice Hall of India, New Delhi, 2005. (Unit I–IV)
2. Mathew M. Radmanesh, "Radio Frequency & Microwave Electronics", Pearson Education Asia, Second Edition, 2002 (Unit – V)
3. Annapurna Das, Sisir K. Das, "Microwave Engineering", McGraw Hill Education (India) private limited, Third edition, 2000. (Unit – V)



Signature of Faculty Member


HOD